PADHAKU
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CHAPTER 8

Playing with Constructions

NCERT Exercise Solutions

8.1 Artwork	5 Q
Construct · 8.1 Artwork	3 Q
Figure it Out · 8.1 Wavy Wave	3 Q
8.2 Squares and Rectangles	2 Q
Figure it Out · 8.2 Squares and Rectangles	4 Q
8.3 Constructing Squares and Rectangles	2 Q
Construct · 8.3 Constructing Squares and Rectangles	3 Q
8.4 An Exploration in Rectangles	6 Q
Construct · Breaking Rectangles	3 Q
Construct · 8.4 An Exploration in Rectangles	6 Q
8.5 Exploring Diagonals of Rectangles and Squares	7 Q

✨ Try each question yourself first, then check the answer. 💡
answers are samples: write your own.
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⭐ 8.1 Artwork

? Observe the way a compass is made. What can one draw with the compass?

✔ ANSWER

A compass has two arms: one with a sharp point (tip) and one that holds a pencil.

Keep the tip fixed and turn the pencil arm around it.

So a compass draws **circles and parts of circles (arcs)**. It can also be used to measure and copy a length.

Think Mark a point P. Imagine marking all the points that are 4 cm away from P. How would they look?

✔ ANSWER

All these points together form a **circle** with centre P and radius 4 cm.

Check: take any point on the curve and measure its distance from P. It is 4 cm.

? How can you mark points 4 cm away from P using a compass?

✔ ANSWER

- Open the compass against a ruler so that the distance between the tip and the pencil is 4 cm.

- Place the tip on P and mark points with the pencil in different directions.
- Keep the tip fixed and move only the pencil all around to get the full curve: a **circle**.

? **Take a point on the circle. What will be its distance from P — equal to 4 cm, less than 4 cm or greater than 4 cm? What about another point on the circle?**

✔ ANSWER

Every point on the circle is **exactly 4 cm** from P (equal to the radius).

? **Recreate the images in Fig. 8.1 with a ruler and compass. Can you make them look as good? Has the use of instruments made the construction easier?**

💡 SAMPLE ANSWER

Draw the straight parts with the ruler and the round parts with the compass.

Yes, the **instruments make it easier**: circles come out perfectly round and lines perfectly straight, which is hard to do freehand.

⭐ Construct · 8.1 Artwork

1 **A Person — How will you draw this? (The figure has two parts: a circle for the head and**

a curve below it.)

💡 SAMPLE ANSWER

- Head: draw a circle with the compass.
- Lower curve: fix a radius in the compass and try placing the tip at different points (above the curve, on the middle line) until the arc passes through both ends of the curve.
- Use your estimate for where to keep the tip, and try again until it looks right (check the picture in your book).

3 **Eyes — How do you draw these eyes with a compass?**

💡 SAMPLE ANSWER

- Draw two light (supporting) horizontal lines for the corners of the eye.
- Upper curve: place the compass tip at a point A below the eye and draw an arc through the two corners.
- Lower curve: place the tip at a point B above the eye, the same distance from the middle as A, and draw the arc through the same two corners.
- If A and B are placed symmetrically, the eye is symmetrical. Draw a small circle inside for the eyeball. Repeat for the second eye (check the picture in your book).

? **Make other artwork of your choice with a ruler and a compass.**

💡 **SAMPLE ANSWER**

Example: a flower — draw a circle, then with the same radius put the tip on the circle and draw arcs around it to get 6 petals.

🧮 Figure it Out · 8.1 Wavy Wave

1 **AB = 8 cm is the central line and the first wave is a half circle from A to X. What radius should be taken in the compass to get this half circle? What should be the length of AX?**

✔ **ANSWER**

There are two equal waves on AB, so X is the middle of AB.

AX = 8 cm ÷ 2 = **4 cm**

The half circle stands on AX, so its radius = 4 cm ÷ 2 = **2 cm**

(Put the compass tip at the middle of AX, 2 cm from A.)

2 **Take a central line of a different length and try to draw the wave on it.**

💡 **SAMPLE ANSWER**

Example: AB = 12 cm.

AX = 12 ÷ 2 = 6 cm, radius = 6 ÷ 2 = 3 cm.

- Draw the first half circle above AX with centre 3 cm from A.
- Draw the second half circle below XB with centre 3 cm from X.

In general, **radius = length of AB ÷ 4**.

3 **Try to recreate the figure where the waves are smaller than a half circle (as in the neck of 'A Person'). Both waves must be identical.**

💡 **SAMPLE ANSWER**

- Choose a radius bigger than half of AX.
- For the first wave, put the compass tip below AB on the line through the middle of AX, at a point from which the arc passes through A and X.
- For the second wave, put the tip above AB, the same distance from the middle of XB, with the same radius.
- Same radius and same distance from the line ⇒ the two waves are identical.

⭐ 8.2 Squares and Rectangles

? **Rectangle ABCD can also be named BCDA, CDAB, DABC, ADCB, DCBA, CBAD and BADC, but not ABDC or ACBD. Can you see what names are allowed and what names are not?**

✔ **ANSWER**

A name is allowed only if the corners come **in order as you travel around the rectangle** (either direction), starting from any corner.

ABDC is not allowed because B to D jumps across the rectangle (B and D are not next to each other).

? **Which of the following is not a name for this square? (Top: S, P; bottom: R, Q)**
1. PQSR 2. SPQR 3. RSPQ 4. QRSP

✔ ANSWER

Going around the square the order is S → P → Q → R.

SPQR, RSPQ and QRSP follow this order.

In PQSR, Q to S jumps across the square.

1. PQSR is not a name for this square.

Figure it Out · 8.2 Squares and Rectangles

1 **Draw the rectangle and four squares configuration (Fig. 8.3) on a dot paper. What did you do so that the four squares are placed symmetrically around the rectangle?**

💡 SAMPLE ANSWER

- Draw the rectangle with its corners on dots.
- At each corner, leave one dot distance diagonally and draw a small square of the same size.

- Place all four squares the same way (same size, same gap) at the four corners, so the figure looks the same on both sides.

2 **Identify if there are any squares in this collection (figures A, B, C, D). Use measurements if needed.**

💡 **SAMPLE ANSWER**

Check each figure: all 4 sides equal and all 4 angles 90°.

Figure A is a square. The others fail one of the properties (check the picture in your book).

Think **Is it possible to reason out if the sides are equal and the angles are right without using any measuring instruments? Can we do this by only looking at the position of corners in the dot grid?**

✔ **ANSWER**

Yes. Count the dots: to go from one corner to the next, count how many dots you move across and how many dots up or down.

If every side has the same ‘across’ and ‘up/down’ counts (just turned around), the sides are equal and the corners are right angles.

3 **Draw at least 3 rotated squares and rectangles on a dot grid with corners on the dots. Verify if they satisfy their properties.**

💡 SAMPLE ANSWER

Example of a rotated square: from a corner go 2 dots right and 1 dot up; next 1 dot left and 2 dots up; next 2 dots left and 1 dot down; then back.

Example of a rotated rectangle: sides of '2 right, 2 up' and '1 left, 1 up'.

Verify: measure the sides (opposite sides equal; for a square all equal) and check each corner is 90° with a set square or protractor.

⭐ 8.3 Constructing Squares and Rectangles

? **Constructing square PQRS of side 6 cm: Can you see why PS should be 6 cm long?**

✔ ANSWER

In a square **all sides are equal** (property S1). PQ = 6 cm, so PS must also be 6 cm.

? **After Step 6: How long is the side RS and what are the measures of ∠R and ∠S?**

✔ ANSWER

RS = 6 cm, **∠R = 90°** and **∠S = 90°**. So the figure is a square.

Construct · 8.3 Constructing Squares and Rectangles

1 **Draw a rectangle with sides of length 4 cm and 6 cm. After drawing, check if it satisfies both the rectangle properties.**

✔ ANSWER

- Draw AB = 6 cm.
- Draw perpendiculars to AB at A and at B.
- With the compass open to 4 cm, mark D on the perpendicular at A and C on the perpendicular at B.
- Join DC.

Check: AB = DC = 6 cm, AD = BC = 4 cm (R1) and ∠A = ∠B = ∠C = ∠D = 90° (R2). **Both properties hold.**

2 **Draw a rectangle of sides 2 cm and 10 cm. After drawing, check if it satisfies both the rectangle properties.**

✔ ANSWER

- Draw PQ = 10 cm.
- Draw perpendiculars at P and Q.
- Mark S and R on them, 2 cm from P and Q. Join SR.

Check: PQ = SR = 10 cm, PS = QR = 2 cm and ∠P = ∠Q = ∠R = ∠S = 90°. **Both properties hold.**

3 **Is it possible to construct a 4-sided figure in which all the angles are equal to 90° but opposite sides are not equal?**

✔ ANSWER

No. If all four angles are 90°, the opposite sides go in the same direction and stay the same distance apart, so they always come out equal. The figure is always a rectangle.

⭐ 8.4 An Exploration in Rectangles

? **Rectangle ABCD with AB = 7 cm and BC = 4 cm; X moves along AD and Y along BC. At which positions will X and Y be at their closest? When will they be farthest?**

✔ ANSWER

- **Closest:** when X and Y are at the same height, i.e. the same distance from A and B (for example X = A and Y = B). Then XY = 7 cm.
- **Farthest:** at opposite corners: X = A and Y = C, or X = D and Y = B. Then XY = diagonal ≈ 8 cm 1 mm.

? **How does the minimum distance between X and Y compare to the length of AB?**

✔ ANSWER

The minimum distance is **equal to AB = 7 cm**.

? **Fill in XY for these positions (shorthand table):**

Distance of X from A	Distance of Y from B	Length of XY
5 mm	3 cm	?
1 cm	1 cm	?
2 cm	4 cm	?

✔ ANSWER

Measured values (about):

Distance of X from A	Distance of Y from B	Length of XY
5 mm	3 cm	**7 cm 4 mm**
1 cm	1 cm	**7 cm 0 mm**
2 cm	4 cm	**7 cm 3 mm**

A table is the shorthand way of recording these.

? **What happens to XY when X and Y are the same distance from A and B?**

Distance of X from A	Distance of Y from B	Length of XY
5 mm	5 mm	?
1 cm	1 cm	?
1 cm 5 mm	1 cm 5 mm	?

✔ ANSWER

Distance of X from A	Distance of Y from B	Length of XY
5 mm	5 mm	**7 cm**
1 cm	1 cm	**7 cm**
1 cm 5 mm	1 cm 5 mm	**7 cm**

1, 2 **In each of these cases, observe (1) how the length XY compares to that of AB and (2) the shape of the 4-sided figure ABYX.**

✔ ANSWER

1. **XY = AB** (7 cm) every time.
2. **ABYX is a rectangle.**

? **How does the farthest distance between X and Y compare with the length of AC? BD?**

✔ ANSWER

The farthest distance is **equal to AC and to BD** (the diagonals). It happens when X and Y are at opposite corners.

⭐ Construct · Breaking Rectangles

Explore **Construct a rectangle that can be divided into two identical squares. If AF = 4 cm, what must the length of AC be? Can the rectangle now be completed?**

✔ ANSWER

AB = AF = 4 cm and BC = AB = 4 cm.

AC = 4 cm + 4 cm = **8 cm**

- Draw AF = 4 cm. Draw a perpendicular to AF at A.
- With the compass open to AF, mark B on it, then C (BC = AB).
- Draw perpendiculars at C and F; they meet at D. Join BE where E is the middle of FD.

Yes, rectangle ACDF (4 cm × 8 cm) is complete.

? **Construct a rectangle that can be divided into three identical squares.**

✔ ANSWER

The length must be **3 times the breadth**.

- Draw the breadth, e.g. 3 cm. Draw a perpendicular at one end.

- With the compass open to 3 cm, mark 3 equal steps on it: length = 9 cm.
- Complete the rectangle (3 cm × 9 cm) and draw lines at each 3 cm mark to show the 3 squares.

? **Give the lengths of the sides of a rectangle that cannot be divided into (a) two identical squares; (b) three identical squares.**

💡 **SAMPLE ANSWER**

(a) Any rectangle whose length is not 2 times its breadth, e.g. **4 cm × 2.5 cm**.

(b) Any rectangle whose length is not 3 times its breadth, e.g. **7 cm × 2 cm**.

⭐ Construct · 8.4 An Exploration in Rectangles

1 **A Square within a Rectangle: Construct a rectangle of sides 8 cm and 4 cm. How will you construct a square inside so that the centre of the square is the same as the centre of the rectangle?**

💡 **SAMPLE ANSWER**

Rough figure: the square touches the top and bottom sides, so its **side = 4 cm**.

Space left = 8 cm − 4 cm = 4 cm, shared equally: 4 ÷ 2 = **2 cm on each side**.

- Draw the 8 cm × 4 cm rectangle.

- On the top and bottom sides mark points 2 cm from each end.
- Join the marks top to bottom. The middle part is a 4 cm square with the same centre (check the picture in your book).

2 **Falling Squares: Construct three squares of side 4 cm aligned as shown. Now try with squares of side 7 cm, 5 cm and 3 cm.**

💡 **SAMPLE ANSWER**

- Draw a horizontal base line and construct the first square on it.
- Each next square starts on the side of the previous square as in the picture; draw its sides with perpendiculars and mark the lengths with the compass.
- For the second figure use sides 7 cm, 5 cm and 3 cm, keeping the same alignment (check the picture in your book).

3 **Shadings: Construct this. Choose measurements of your choice. The larger 4-sided figure is a square and so are the smaller ones.**

💡 **SAMPLE ANSWER**

Example: draw a square of side 8 cm. Mark the midpoints of the sides (4 cm) and draw lines through them to get 4 smaller squares of side 4 cm. Shade the parts as in the picture (check the picture in your book).

4 **Square with a Hole: The centre of the circular hole is the same as the centre of the square. Where should the centre of the circle be?**

✔ ANSWER

- Construct the square.
- Join its diagonals (opposite corners). They meet at the centre of the square.
- Put the compass tip at this meeting point and draw the circle.

So the centre of the circle is **where the diagonals of the square meet**.

5 **Square with more Holes: Construct the figure.**

💡 SAMPLE ANSWER

- Construct the square and divide it into smaller equal squares (join midpoints of opposite sides).
- Find the centre of each small square by joining its diagonals.
- Draw a circle of the same radius at each centre (check the picture in your book).

6 **Square with Curves: a square with 8 cm sides. Where can the tip of the compass be placed so that all 4 arcs bulge uniformly from each side?**

💡 **SAMPLE ANSWER**

- Construct the 8 cm square.
- For each side, put the compass tip on the line through the centre of the square that is perpendicular to that side (for example at the midpoint of the opposite side).
- Open the compass to reach the two corners of the side and draw the arc between them.
- Use the same position and the same radius for all 4 sides, so all arcs bulge equally (check the picture in your book).

⭐ 8.5 Exploring Diagonals of Rectangles and Squares

? **Consider a rectangle PQRS. Join PR and QS (the diagonals). Compare the lengths of the diagonals.**

✔ **ANSWER**

On measuring, **PR = QS**. The two diagonals of a rectangle are equal.

? **Diagonal PR divides ∠R into g and h, and ∠P into c and d. Are g and h equal? Are c and d equal? Identify pairs of angles that are equal.**

💡 SAMPLE ANSWER

In a rectangle that is not a square, **g ≠ h and c ≠ d** (but c + d = 90° and g + h = 90°).

Equal pairs: **c = g** and **d = h**; on diagonal QS, **e = a** and **f = b**.

Also d = e = a = h and c = f = b = g (the angles the diagonals make with the long sides are equal, and so are those with the short sides) (check the picture in your book).

Explore **How should the rectangle be constructed so that the diagonal divides the opposite angles into equal parts?**

✔ ANSWER

Make the two adjacent sides **equal**, i.e. construct a **square**. Then each diagonal divides the 90° angles into 45° and 45°.

? **Did you consider the case of a square? See what happens in this special case!**

✔ ANSWER

In a square, all 8 angles made by the diagonals are **45°**, and the diagonals are still equal.

Math Talk **What general laws did you observe with respect to the angles and sides? How can one be sure if the laws will always be true?**

💡 **SAMPLE ANSWER**

- The two diagonals of a rectangle are equal.
- The two angles at each corner add up to 90°.
- The diagonals divide the angles equally (45° each) only when the rectangle is a square.

We can be sure only by **reasoning (proof)** that works for every rectangle; measuring a few rectangles only gives us confidence.

? **Worked example (60° and 30°): Draw AB, the perpendicular at B and ∠BAC = 60°. A line through A perpendicular to AB divides ∠A into two angles. One measures 60°. Check what the other angle is.**

✔ **ANSWER**

∠A = 90°, so the other angle = 90° − 60° = **30°**.

? **Worked example (side 5 cm, diagonal 7 cm): Consider the point where the circle (centre D, radius 7 cm) and the line l intersect. What is its distance from point D?**

✔ **ANSWER**

Its distance from D is **7 cm**, because every point of the circle is 7 cm from D. So this point is B.

⭐ Construct · 8.5 Exploring Diagonals

1 **Construct a rectangle in which one of the diagonals divides the opposite angles into 50° and 40°.**

✔ ANSWER

- Draw AB of any length, e.g. 6 cm.
- At B draw a line perpendicular to AB.
- At A draw ∠BAC = 40° with a protractor; its arm meets the perpendicular at C.
- Draw a line through A perpendicular to AB. Now ∠CAD = 90° − 40° = 50°.
- Mark D with the compass so that AD = BC (or draw a perpendicular to BC at C). Join CD.

ABCD is the rectangle; diagonal AC divides ∠A and ∠C into **40° and 50°**.

2 **Construct a rectangle in which one of the diagonals divides the opposite angles into 45° and 45°. What do you observe about the sides?**

✔ ANSWER

- Draw AB, the perpendicular at B, and ∠BAC = 45°; C is where the arm meets the perpendicular.
- Complete the rectangle as in Q1.

Observation: **AB = BC, so all four sides are equal**. The rectangle is a **square**.

3 **Construct a rectangle one of whose sides is 4 cm and the diagonal is of length 8 cm.**

✔ ANSWER

- Draw DC = 4 cm.
- Draw line l perpendicular to DC at C.
- With centre D and radius 8 cm, draw an arc cutting l at B (DB = 8 cm).
- Draw perpendiculars to DC at D and to BC at B; they meet at A.

ABCD is the rectangle. (BC comes out about 6.9 cm.)

4 **Construct a rectangle one of whose sides is 3 cm and the diagonal is of length 7 cm.**

✔ ANSWER

- Draw DC = 3 cm.
- Draw line l perpendicular to DC at C.
- With centre D and radius 7 cm, draw an arc cutting l at B.
- Draw perpendiculars at D (to DC) and at B (to BC); they meet at A.

ABCD is the rectangle. (BC comes out about 6.3 cm.)

⭐ 8.6 Points Equidistant from Two Given Points

? **House: The circles of radius 5 cm with centres B and C meet at a point. How far is it from B? How far is it from C?**

✔ ANSWER

It is on both circles, so it is **5 cm from B** and **5 cm from C**. This point is A.

Think **Was it necessary to draw two full circles to get the point A?**

✔ ANSWER

No. Only the parts near A are needed, so two small arcs of radius 5 cm from B and C are enough.

? **Having obtained point A, how do we draw the remaining arc (the roof curve)?**

✔ ANSWER

A is 5 cm from both B and C, so put the compass tip at A with radius 5 cm and draw the arc from B to C.

⭐ Construct · 8.6 Points Equidistant from Two Given Points

1 **Construct a bigger house in which all the sides are of length 7 cm.**

✔ ANSWER

- Construct square BCED of side 7 cm (DE = 7 cm, perpendiculars at D and E, BD = CE = 7 cm, join BC).
- Draw the door (e.g. 1 cm × 2 cm) on DE.
- With radius 7 cm draw arcs from B and from C; they meet at A. Join AB and AC.
- With centre A and radius 7 cm, draw the arc from B to C.

2 **Try to recreate ‘A Person’, ‘Wavy Wave’ and ‘Eyes’ from the section ‘Artwork’, using ideas involved in the ‘House’ construction.**

💡 SAMPLE ANSWER

Idea from the House: the compass tip for an arc through two points must be the **same distance from both points**. Find it by drawing arcs of equal radius from the two end points; where they cross is the tip position.

- Wavy Wave: arcs of equal radius from A and X (below AB) give the tip for the first wave; from X and B (above AB) for the second.

- A Person and Eyes: do the same with the two end points of each curve. Equal radius on both sides keeps the curves identical and symmetrical.

3 **Is there a 4-sided figure in which all the sides are equal in length but is not a square? If such a figure exists, can you construct it?**

✔ ANSWER

Yes. For example, take all sides 5 cm.

- Draw AC = 5 cm. Draw AB = 5 cm at an angle that is not 90° (e.g. 60°).
- With radius 5 cm draw arcs from B and from C; they meet at D.
- Join BD and CD. All four sides are 5 cm but the angles are not 90°, so **ABDC is not a square** (it is called a rhombus).

🎉 **Chapter 8 done! Revise the answers once more tomorrow.**